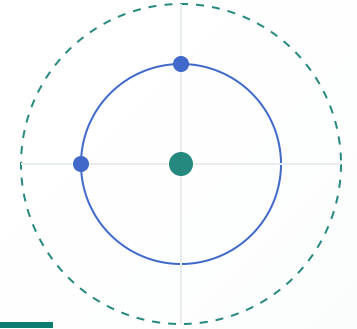


MediSync AI

AI-Powered Triage Chatbot, Prescription Digitizer & Virtual Consultation Hub



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COURSE & PROPOSAL

CSE327 Software Engineering
Project Proposal Presentation
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The Problem

REAL WORLD CLINICAL FRICTION & IMPACT METRICS

54%

Drugstore Reliance

In Bangladesh, **54%** of patients rely directly on unqualified drugstore sellers for primary healthcare advice due to lack of accessible clinics.

9.9

Workforce Deficit

The formal healthcare workforce is critically low in Bangladesh, with only **9.9** professionals available per 10,000 people.

18M

Global Shortage

Driven by a worldwide shortage of **18 million** health workers, informal providers manage up to **55%–77%** of primary care across low-income regions.

The Solution

COMPREHENSIVE FUNCTIONAL CORE BLUEPRINT

1. AI Medical Consultation

Provides instant symptom guidance to help patients safely, replacing the risky reliance on unqualified drugstore sellers.

2. Prescription Analyzer

Uses Gemini AI to read prescription photos, explain medicine details simply, and show cheaper generic alternatives.

3. Medicine Taking Reminders

Sends automatic notification alarms to the patient's phone to make sure they take their medicines on time.

4. Pharmacy Stock Chat

Shows if medicines are available nearby and lets patients chat directly with pharmacy sales reps to verify stock.

The Roadmap

PROJECT DEVELOPMENT LIFE CYCLE



1. DESIGN PHASE

Architecture Design



DB Schema Specs



UI/UX Wireframing



2. CORE CODING

Mobile App Shell



Web Portals (Doctor/Pharmacy)



Express API Routes



3. AI & FEATURES

Gemini Consultation Chat



Multimodal Prescription OCR



Stock & Reminders



4. TESTING & DEPLOY

Chat & Alarm Verification



Security Audit



Cloud Deploy & B2B Ads

Types of Users

ACCESS BOUNDARIES & DEVICE LAYOUTS

App Users (Who Uses the Platform)

Patient

Checks symptoms via AI, consults real doctors virtually, and receives pill alarm notifications.

Doctor

Conducts virtual consultations and issues digital prescription reports directly to patients.

Pharmacy Salesperson

Manages stock status, chats directly with patients, and updates drug availability.

Device Interfaces (How They Access It)

Mobile Application

Convenient smartphone app built for patients to consult, scan prescription logs, and get reminders.

Web Portal

Professional web dashboards built for doctors to manage consults and pharmacy sales reps to chat.

Functional Requirements

CORE SYSTEM CAPABILITIES (FR)

FR1 – AI Medical Consultation

Conversational health chatbot that queries symptom logs to guide patients and recommend clinical doctors.

FR2 – Prescription Analyzer

Scans scripts via Gemini API to extract details, explain dosages, and list generic alternatives.

FR3 – Notification Reminders

Allows setting alarm triggers on patient devices to send automatic notifications for medicine schedules.

FR4 – Pharmacy Map & Live Chat

Maps pharmacy locations to list stock availability and enables direct live chat with salespersons.

Non-Functional Requirements

SECURITY, SPEED & RELIABILITY TARGETS

Simple Security & Privacy

User passwords will be hashed securely, and patient data will be stored privately without sharing with third parties or public networks.

Responsive Performance

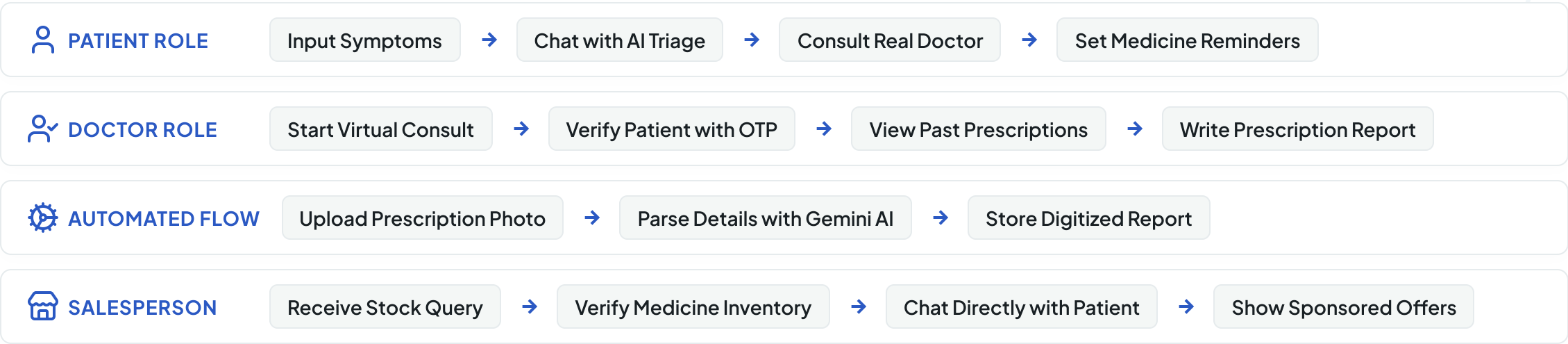
The AI consultation chat and prescription scanner will respond within 4–5 seconds, keeping the patient experience smooth and interactive.

Standard Availability

The application server will maintain standard web hosting uptime, ensuring the system is available whenever users need to access stock checks or triage.

System Access & Role Specific Workflows

HOW USERS INTERACT WITH THE APP



User Stories & Acceptance Criteria

ENGINEERING VALIDATION FRAMEWORK

Consumer: Patient

"As a patient, I want to scan prescriptions, set pill alarms, and chat with an AI doctor so I can recover on a budget."

- ✓ **AC 1:** Image scanner outputs drug details and cheaper generic alternatives.
- ✓ **AC 2:** Schedules push notification alarms on patient mobile device.

Professional: Doctor

"As a physician, I want to consult virtually and write digital prescription reports so I can treat patients online."

- **AC 1:** Dashboard enables video/chat consults and PDF report generation.
- **AC 2:** Requires patient OTP validation before showing historical logs.

Business: Pharmacy Rep

"As a salesperson, I want to verify medicine availability via chat and promote my store to nearby patients."

- ✓ **AC 1:** Connects salesperson and patient in direct stock check chat.
- ✓ **AC 2:** Enables B2B ad prioritization placement for availability list.

DISTRIBUTED SERVICES TOPOGRAPHY BLUEPRINT



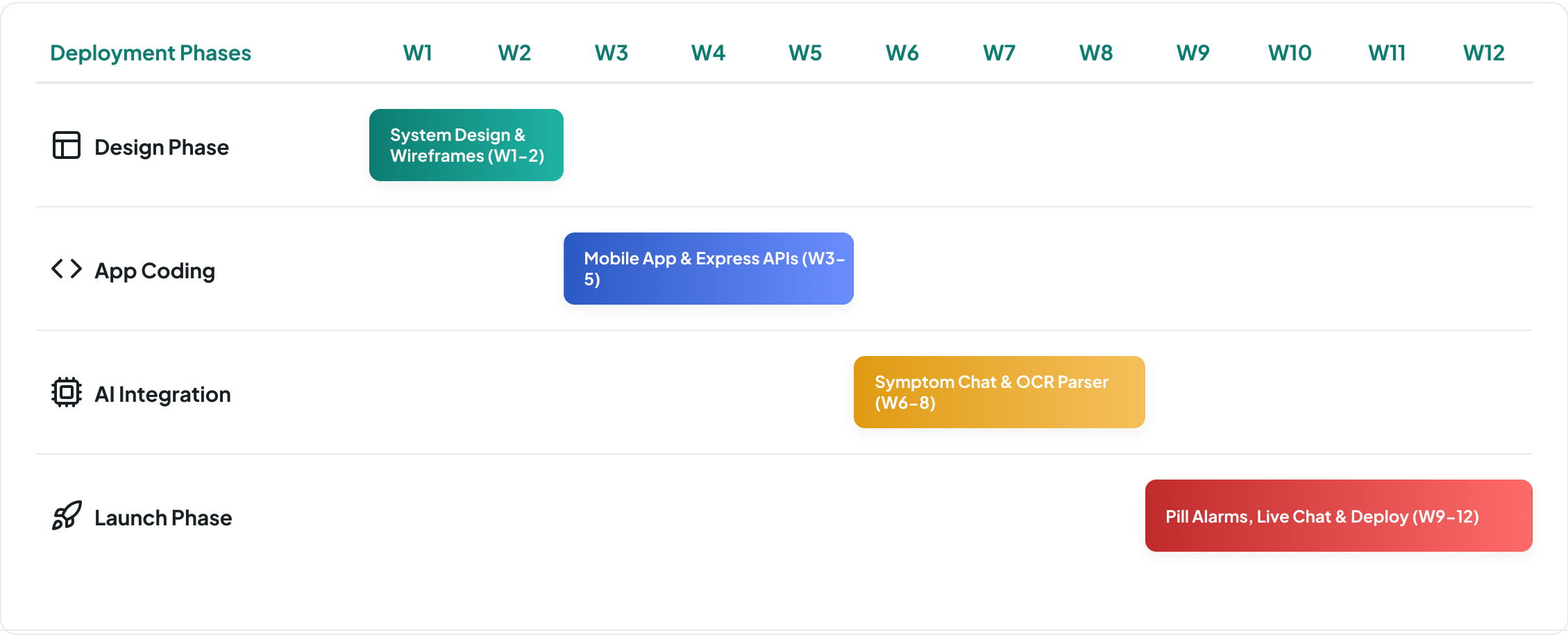
Infrastructure & Technology Stack

INTEGRATED SYSTEM COMPONENTS

Architectural Tier	Technology Selection	Core Operational Intent
Client Core	React Native / Flutter Frameworks	Cross-platform mobile access deploying location and camera assets
Clinic Hub	React.js + Tailwind CSS Utility Engine	Responsive desktop rendering for complex multi-column health analytics
Backend Gateway	Node.js + Express Server Runtime	High-performance routing layer managing profile transactions
AI Intelligence	Gemini API Integration	Executes prescription OCR analysis, symptom triage chatbot, and generic matching
Data Integration	PostgreSQL + Secure Local Storage	Manages accounts and locks upload folders using session-based login validation

Project Timeline

12-WEEK AGILE DEPLOYMENT SCHEDULE



Market Precedence & Reference Systems

COMPARATIVE INDUSTRY LANDSCAPES

Reference App: DocTime Platform

DocTime is a popular app providing virtual doctor consultations across multiple medical specializations for low, affordable fees.

What DocTime Lacks (Our Focus):

It acts purely as a consultation booking portal. It lacks a conversational AI symptom advisor, does not scan physical prescriptions to suggest cheaper generic drugs, and has no direct stock check connection with pharmacy sales representatives.

The MediSync AI Solution Shift

MediSync AI wraps virtual doctor consultations inside a complete patient helper. Before a doctor consult, the AI checks symptoms. After the consult, the app scans the doctor's prescription report to offer cheaper alternatives, sets alarm notifications for pill intake, and lets patients chat with pharmacists to verify stock availability.

Proposed Novelty

ARCHITECTURAL DIFFERENTIATION ADVANTAGES

1. Conversational Triage

Replaces typical search fields with an active LLM interface that dynamically asks follow-up questions to route users accurately.

2. Vision Extraction

Leverages Gemini API to instantly structure handwritten script images into digital medicine logs, bypassing manual model training.

3. Market Transparency

Resolves medical cost barriers by analyzing active ingredient salts to automatically match patients with affordable options.

Feasibility Analysis

TECHNICAL & PRACTICAL OPERATIONAL VIABILITY

✓ Technical: Highly Feasible

Yes, we can build this. By calling Gemini API directly for AI consults and OCR instead of training models, the system is lightweight and buildable.

✓ Operational: Highly Feasible

Yes, we can build this. The app relies on standard web interfaces (React & Node.js) that we know how to code, requiring no complex hospital integrations.

✓ Financial: Highly Feasible

Yes, we can build this. The database and server runtime will host entirely on free cloud servers, keeping our development cost at zero.

⚠ Safety Challenge: Difficult

Accuracy is difficult. Scans of handwritten prescriptions are not 100% correct. Since medicine is sensitive, we will require user verification and display safety warnings before acting.

Impact of the Program

SOCIO-ECONOMIC & CLINICAL HEALTH ADVANCES

Overcoming Doctor Shortages

Provides safe, instant health guidance to the 54% of patients relying on unqualified drugstores, reducing clinic burden.

Lowering Medicine Costs

Exposes cheaper generic drug alternatives directly to patients, helping low-income families afford full treatments.

Improving Pill Adherence

Reduces missed medication doses by sending automatic alarm notifications directly to patients' phones.

Reducing Drug Access Time

Saves hours of search time by mapping local pharmacies and hosting live chats to check medicine stock.

Thank You

Questions & Discussion

